

Heterogeneous Active Retrieval Across Structurally Different Evidence Substrates

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Abstract

FLARE and Self-RAG establish important forms of adaptive generation-time retrieval [1, 2]. This paper does not claim that mechanism. It tests whether structurally different information needs benefit from routing among genuinely heterogeneous evidence substrates: relational execution, lexical indexing, vector retrieval, web search, and optionally graph traversal. The proposed router remains transparent and heuristic. The decisive controls are matched-cost universal retrieval, broadcast, explicit model selection, and oracle source routing. If oracle heterogeneous routing cannot beat a universal retriever, later learned routing is not motivated.

1 Architecture

The runtime forms a typed information-need event

$$h_t \rightarrow e_t = (\text{entity, relation, time, source hints, confidence}),$$

then a transparent router chooses among

$$\{\text{DB, lexical, vector, KG, web}\}.$$

Each event is logged through the factorization

$$\text{TRIGGER} \rightarrow \text{SOURCE} \rightarrow \text{QUERY} \rightarrow \text{RETRIEVE/EXECUTE} \rightarrow \text{EVIDENCE USE}.$$

Component scores and oracle substitutions prevent routing failures from being hidden inside final-answer accuracy.

2 Minimum Retrieval Suite

- relational/structured database;
- lexical/document index;
- vector database or semantic document retriever;
- web/search source.

A knowledge graph is optional if it adds a genuinely distinct access pattern. Multiple wrappers over one text retriever do not satisfy this requirement.

3 Heuristic Routing

Examples:

- structured attributes, identifiers, aggregates $\rightarrow$ DB;
- explicit document names/exact terms $\rightarrow$ lexical;
- semantic document questions $\rightarrow$ vector;
- latest/current/recent external facts $\rightarrow$ web;
- entity/relation neighborhoods $\rightarrow$ KG.

The benchmark makes these semantics necessary: an argmax over sales requires DB aggregation, an exact policy clause favors lexical lookup, a semantic report question favors vector retrieval, a current public fact favors web search, and a neighborhood or path query favors a KG. No learned router is allowed in the core paper.

4 Composed Helpers

Small deterministic preprocessing may resolve relative dates, normalize entity IDs, or form database filters/aggregations. Bounded compositions such as date resolver $\rightarrow$ web and entity resolver $\rightarrow$ DB are allowed, but their dependency DAG and component costs must be logged. They may not become a hidden general planner.

5 Common Evidence IR

All retrievers emit source type/ID, query, content/value, timestamp/freshness, relevance score where meaningful, support/contradiction/ambiguity status, and provenance.

6 Tasks

Use source-diverse workloads: structured business facts, exact document lookup, semantic document QA, recent/current facts, multi-source conflicts, no-retrieval cases, and cases where the best source becomes clear only mid-generation.

7 Baselines

1. LLM only.
2. Single universal retriever.
3. Upfront RAG across all sources.
4. FLARE-like confidence trigger plus universal retriever.
5. FLARE-like confidence trigger plus heuristic heterogeneous router.
6. Self-RAG-inspired adaptive retrieval, if feasible and clearly labeled as an approximation unless canonical training is reproduced.

7. Explicit LLM tool/source selection.
8. Broadcast-to-all under matched cost and fanout.
9. Heuristic semantic reflex routing.
10. Oracle source router.
11. Oracle trigger plus source router.

8 Metrics

Report final accuracy and unsupported claims, plus separate trigger P/R, source top-1/top- k and confusion matrix, query correctness, retrieval/execution P/R, evidence-use rate, source fanout, conflicts, cost, generated/tool-selection tokens, latency, dependency-DAG cost, and scaling with source count and factual subgoals.

9 Falsification

The heterogeneous claim fails if a universal retriever matches under equal cost, oracle routing adds little, matched-cost broadcasting dominates, source-specific query semantics do not matter, explicit source selection is equally efficient, or routing errors dominate specialization gains.

10 Scope

No learned routing, hidden-state trigger, backtracking, or PRA/native KV integration.

References

- [1] Z. Jiang et al. Active retrieval augmented generation. *EMNLP*, 2023. <https://arxiv.org/abs/2305.06983>.
- [2] A. Asai et al. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. *ICLR*, 2024. <https://arxiv.org/abs/2310.11511>.